Asim Mohamed

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Research interests

Trustworthy and secure AI, content provenance and LLM watermarking, machine learning auditing, adversarial machine learning, multi-agent safety, interpretability and explainability.

Education

MSc Machine Intelligence, African Institute for Mathematical Sciences Feb 2024 – Jan 2026
African Master's in Machine Intelligence (AMMI). Fully funded by Google and Meta AI.

- Thesis: Is Multilingual LLM Watermarking Truly Multilingual? A Simple Back-Translation Solution. Supervisor: Dr Martin Gubri - Assistant Professor at École Polytechnique. [pdf] [code]
- Coursework: computational linear algebra, mathematics for ML, foundations of ML and deep learning, optimisation for ML, NLP, computer vision, MLOps.

BSc (Hons) Electrical and Electronic Engineering, University of Khartoum Oct 2015 – Dec 2021
Specialisation in electronic systems software engineering. First class honours, CGPA 7.93/10.

- Thesis: Prediction of Regional Biocapacity Using Satellite Imagery and Deep Learning. Supervisor: Dr Anas Showk. [pdf] [code]
- Coursework: data structures and algorithms, neural networks and fuzzy logic, distributed and parallel computing, AI and expert systems, databases, computer networks.

Publications

Peer-reviewed

[1] **Asim Mohamed** and Martin Gubri. *Is Multilingual LLM Watermarking Truly Multilingual? Scaling Robustness to 100+ Languages via Back-Translation.* The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026). [arXiv] [code]

[2] Ahmed Alhassan, **Asim Mohamed** and Moayad Elamin. *Phantoms at BAREC Shared Task 2025: Enhancing Arabic Readability Prediction with Hybrid BERT and Linguistic Features.* Proceedings of the Third Arabic NLP Conference (ArabicNLP 2025), shared tasks. [paper]

[3] Muhammed Saeed, **Asim Mohamed**, Mukhtar Mohamed, Shady Shehata and Muhammad Abdul-Mageed. *From Nile Sands to Digital Hands: Machine Translation of Coptic Texts.* Proceedings of the Second Arabic NLP Conference (ArabicNLP 2024), pp. 298–308. [paper]

Research projects

STEAM: translation-robust watermark detection MSc thesis

- Showed that existing multilingual watermarking methods lose robustness under translation attacks in medium- and low-resource languages. Traced the failure to semantic clustering, which breaks when the tokeniser has too few full-word tokens for a language.
- Proposed STEAM, a detection method that uses Bayesian optimisation to search over 100+ candidate languages for the back-translation that best recovers watermark strength. It works with any watermarking scheme and adds no cost at generation time.

Kinyarwanda ML evaluation platform In progress

- Built a benchmark and public leaderboard that compares Kinyarwanda ASR, TTS and LLM models. ASR is scored automatically with WER and CER. A panel of human raters scores TTS and LLM outputs, with hidden gold items and inter-rater agreement checks for quality control. Teams submit model predictions and receive a ranked, published result.

Towards Robust and Safe AI: A Comprehensive Survey In progress

- Conducted an in-depth review of adversarial vulnerabilities in multilingual and low-resource language models, classifying attack methods and analysing their implications for model safety and reliability.
- Examined the impact of prompt injection attacks on the robustness of multilingual LLMs, focusing on their ability to bypass safety mechanisms and generate harmful or unintended outputs under adversarial conditions.

Industry experience

Senior AI engineer, Manzilo Apr 2026 – present

- Architected a multi-tenant AI lead-acquisition platform with structural tenant isolation and role-based access control.
- Designed five LangGraph agents for lead extraction, enrichment, scoring, outreach and live conversation. Added multi-model orchestration, queue-based retries and dead-letter recovery.

Enigma-AI Feb 2023 – present

Technical lead (2026 – present) · AI engineer (2025 – 2026) · Software engineer (2023 – 2024)

- Lead a team of four engineers. Own system architecture, technical standards and hiring for an agent platform serving 12+ client applications.
- Built an agent evaluation system with automated quality scoring and cost tracking. Reduced response latency by 43% and token cost by 16% across production agents.
- Led retrieval-quality and Sudanese Arabic dialect-tuning work that improved answer accuracy and relevance by 31%.
- Built a hybrid dense and sparse retrieval RAG pipeline with ingestion from PDF, DOCX, websites and scanned documents.

Full-stack engineer, Malomatia May 2022 – Apr 2023

- Modernised e-services for Qatar's Supreme Judiciary Council by migrating from ASP.NET Web Forms to React and TypeScript.

Teaching and talks

Workshop instructor, IndabaX Sudan 2025 (remote) Dec 2025

- Practical workshop on LLM foundations: tokenisation, embeddings and attention in transformers. [materials]

Poster, Deep Learning Indaba 2024, Dakar Sep 2024

- Presented the first Coptic-centred parallel corpus and an evaluation of LLMs translating between Coptic, English and Arabic [3]. [poster]

Lab instructor, White Nile University Oct 2023 – Jan 2024

Algorithms and data structures; multimedia technology and applications.

Lab instructor, University of Khartoum Sep 2022 – Dec 2022

Multimedia technology and applications; networked multimedia.

Honours and schools

- AMMI scholarship, fully funded by Google and Meta AI 2024
- First class honours, BSc, University of Khartoum 2021
- MenaML Winter School, KAUST, Saudi Arabia Jan 2026
- Deep Learning Indaba, Dakar and Kigali 2024, 2025
- Cohere Labs Open Science Community Summer School Jul 2025

Skills and communities

ML research	PyTorch, JAX/Flax, TensorFlow/Keras, Hugging Face, scikit-learn, NumPy, pandas, OpenCV
LLM systems	LangGraph, LangChain, RAG, Qdrant, Chroma, MCP, OpenAI, , OpenRouter, Anthropic and Gemini APIs
Engineering	Python, TypeScript, JavaScript, C#, SQL, Java, Dart; FastAPI, Node.js, Nest.js, React.js, Next.js; Docker, AWS, Google Cloud, Redis, PostgreSQL, MongoDB, Convex
Communities	Black in AI, Masakhane, Cohere Labs Open Science Community, EleutherAI, Deep Learning Indaba, Sudanese Machine Learning Community, MenaML

References

Dr Martin Gubri
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Member of technical staff, Anthropic.
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Dr Anas Showk
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